

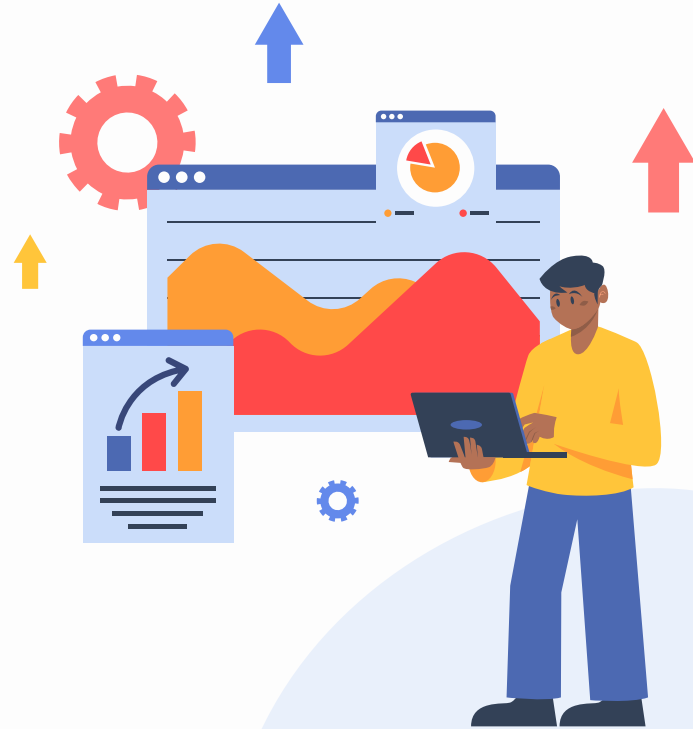
Muhamad Fajar Suryadi

Netflix Data Analysis

Exploratory Data Analysis & Business Insight

Link Dashboard :

<https://lookerstudio.google.com/reporting/4a6b838c-1a1d-4f8d-9438-e28cddf931ec>





Data Cleaning and Process

Data Pipeline

<u>Google Sheets</u>	Import dataset CSV awal
<u>BigQuery</u>	Data Cleaning & Transformation (SQL)
<u>Looker Studio</u>	Visualisasi & Dashboard
<u>Google Slides</u>	Presentasi hasil analisis

Google Sheets → BigQuery → Looker Studio → Google Slides





Google Sheets Import Data

netflix_titles ☆ 📁 ☁

File Edit View Insert Format Data Tools Extensions Help

100% \$ % .0 .00 123 Default... - 10 + B I A

L20 After a deadly home invasion at a couple's new dream house, the traumatized wife searches for answers — and learns the real danger is just beginning.

	A	B	C	D	E	F	G	H	I	J	K	L
	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
1	s1	Movie	Dick Johnson Is	Kirsten Johnson		United States	September 25, 2	2020	PG-13	90 min	Documentaries	As her father nears the
2	s2	TV Show	Blood & Water		Ama Qamata, KI	South Africa	September 24, 2	2021	TV-MA	2 Seasons	International TV	After crossing paths at
3	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, San		September 24, 2	2021	TV-MA	1 Season	Crime TV Shows	To protect his family fro
4	s4	TV Show	Jailbirds New Orleans				September 24, 2	2021	TV-MA	1 Season	Docuseries, Rea Feuds, flirtations and to	
5	s5	TV Show	Kota Factory		Mayur More, Jite	India	September 24, 2	2021	TV-MA	2 Seasons	International TV	In a city of coaching cer
6	s6	TV Show	Midnight Mass	Mike Flanagan	Kate Siegel, Zach Gilford, Hamish		September 24, 2	2021	TV-MA	1 Season	TV Dramas, TV	The arrival of a charism
7	s7	Movie	My Little Pony: A	Robert Cullen, J	Vanessa Hudgens, Kimiko Glenn,		September 24, 2	2021	PG	91 min	Children & Famil	Equestria's divided. But
8	s8	Movie	Sankofa	Haile Gerima	Kofi Ghanaba, C	United States, G	September 24, 2	1993	TV-MA	125 min	Dramas, Indepei	On a photo shoot in Gh
9	s9	TV Show	The Great Britis	Andy Devonshir	Mel Giedroyc, Si	United Kingdom	September 24, 2	2021	TV-14	9 Seasons	British TV Show	A talented batch of ame
10	s10	Movie	The Starling	Theodore Melfi	Melissa McCarti	United States	September 24, 2	2021	PG-13	104 min	Comedies, Dram	A woman adjusting to li
11	s11	TV Show	Vendetta: Truth, Lies and The Mafia				September 24, 2	2021	TV-MA	1 Season	Crime TV Shows	Sicily boasts a bold "An
12	s12	TV Show	Bangkok Breakir	Kongkiat Komes	Sukollawat Kanarot, Sushar Man		September 23, 2	2021	TV-MA	1 Season	Crime TV Shows	Struggling to earn a livi
13	s13	Movie	Je Suis Karl	Christian Schwo	Luna Wedler, Ja	Germany, Czech	September 23, 2	2021	TV-MA	127 min	Dramas, Internat	After most of her family
14	s14	Movie	Confessions of a	Bruno Garotti	Klara Castanho, Lucca Picon, J		September 22, 2	2021	TV-PG	91 min	Children & Famil	When the clever but so
15	s15	TV Show	Crime Stories: India Detectives				September 22, 2	2021	TV-MA	1 Season	British TV Show	Cameras following Ben
16	s16	TV Show	Dear White People		Logan Browning	United States	September 22, 2	2021	TV-MA	4 Seasons	TV Comedies, T	Students of color navig
17	s17	Movie	Europe's Most D	Pedro de Echave	García, Pablo Azorín Williams		September 22, 2	2020	TV-MA	67 min	Documentaries	Declassified documents
18	s18	TV Show	Falsa identidad		Luis Ernesto Fra	Mexico	September 22, 2	2020	TV-MA	2 Seasons	Crime TV Shows	Strangers Diego and Is
19	s19	Movie	Intrusion	Adam Salky	Freida Pinto, Logan Marshall-Gre		September 22, 2	2021	TV-14	94 min	Thrillers	After a deadly home inv

+ ≡ raw_data ▾

Dataset CSV di-import ke Google Sheets sebagai tahap awal eksplorasi dan validasi struktur data sebelum diproses di BigQuery.





Bigquery Import

Create table

Source

Create table from
Upload

Select file *
netflix_clean_sheet.csv

File format
CSV

Destination

Project *
netflixanalysis-492718

Dataset *
netflix_ds

Table *
netflix_clean_sheets

Maximum name size is 1,024 UTF-8 bytes. Unicode letters, marks, numbers, connectors, dashes, and spaces are allowed.

Table type
Native table

Create table Cancel

netflixanalysis-492718 / Datasets / netflix_ds / Tables / netflix_clean_sheet

☆ netflix_clean_... Query Open in Share Copy Paste Delete Refresh

Schema Details Preview Table Explorer Preview Insights Lineage Data Profile Data Quality

Row	show_id	type	title	director	cast
1	s6785	Movie	Forbidden Planet	Fred M. Wilcox	Leslie Nielsen Anne Francis Warren Stevens Earl Holliman
2	s3545	Movie	Spookley the Square Pumpkin	Bernie Denk	Sonja Ball, Karen Jones, Michael Gauthier-Francois Groulx, Terrance Bruce Dinsdale

Results per page: 200 1 - 200 of 8807

Job history

Data dari Google Sheets di-import ke BigQuery untuk proses cleaning dan transformasi menggunakan SQL.





SQL Cleanning di Bigquery



```
Untitled query [Run] [Save] [Download] [Share]

1 SELECT
2   show_id,
3   type,
4   title,
5   country,
6   CAST(release_year AS INT64) AS release_year,
7   rating,
8
9   CASE
10    WHEN type = 'Movie'
11    THEN SAFE_CAST(REGEXP_EXTRACT(duration, r'(\d+)') AS INT64)
12  END AS duration_minutes,
13
14  CASE
15    WHEN type = 'TV Show'
16    THEN SAFE_CAST(REGEXP_EXTRACT(duration, r'(\d+)') AS INT64)
17  END AS season_count
18
19 FROM `netflixanalysis-492718.netflix_ds.netflix_clean_sheett`

✓ Query completed
Using on-demand processing quota
```

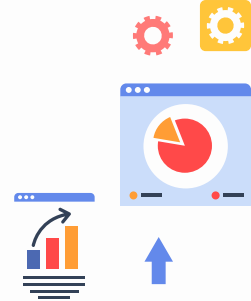
SQL Cleanning di Bigquery

Query results

+ Create conversation

 Save results

Job information		Results	Visualization	JSON	Execution details		Execution graph	
Row	show_id	type	title	country	rele...	rating	duration_min...	season_...
1	s6828	TV Show	Gargantia on th...	Japan	2013	null	null	1
2	s7538	Movie	My Honor Was ...	Italy	2015	null	115	null
3	s7313	TV Show	Little Lunch	Australia	2015	null	null	1
4	s5990	Movie	13TH: A Conver...	null	2017	null	37	null
5	s5814	Movie	Louis C.K.: Live ...	United States	2015	66 min	null	null
6	s5542	Movie	Louis C.K. 2017	United States	2017	74 min	null	null
7	s5795	Movie	Louis C.K.: Hilari...	United States	2010	84 min	null	null
8	s6785	Movie	Forbidden Planet	United States	1956	G	98	null





Create View



Save view



The destination dataset for a saved view must be in the same region as the source, otherwise a "Dataset not found" error will be returned.

Project *

netflixanalysis-492718

Browse

Dataset *

netflix_ds

Table *

analysis_netflix_view

Maximum name size is 1,024 UTF-8 bytes. Unicode letters, marks, numbers, connectors, dashes, and spaces are allowed. The job will create the specified destination table if needed.

SAVE

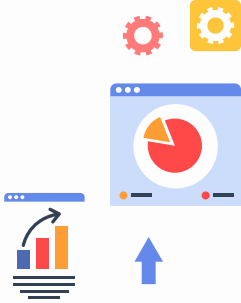
Cancel

Hasil cleaning disimpan sebagai View di BigQuery untuk digunakan langsung pada Looker Studio.

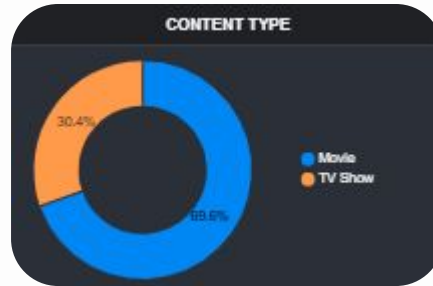
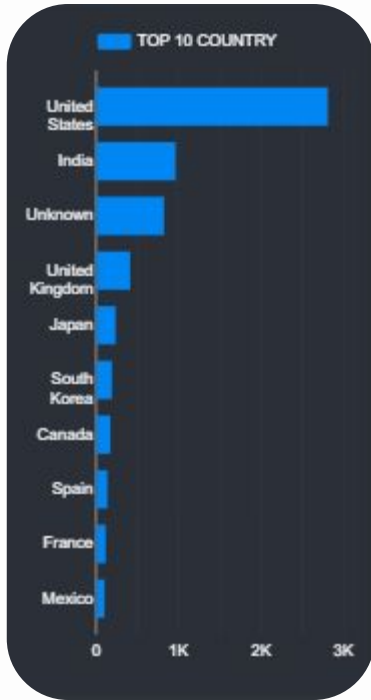
Dashboard Overview



Dashboard
dibuat
menggunakan
Looker Studio
untuk
memvisualisak
an hasil analisis
secara
interaktif.



Movie vs Tv Show dan Top 10 Country

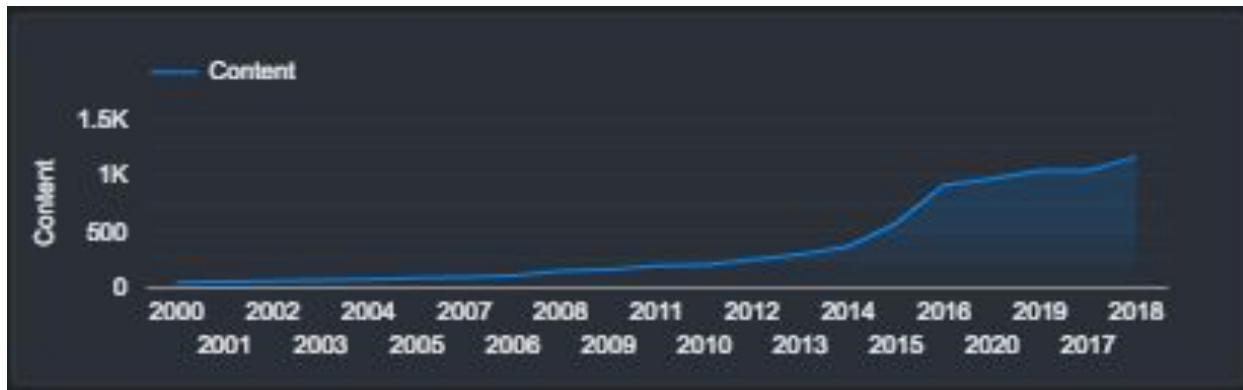


Dominasi Movie 69,6% berdurasi tunggal menunjukkan preferensi konsumsi cepat dibanding long-form content, yang relevan untuk strategi engagement platform. Dan United States menjadi kontributor terbesar konten Netflix.





Release Year



Jumlah konten meningkat signifikan setelah tahun 2015





Duration



Mayoritas Movie berdurasi 90-120 menit dan Mayoritas TV Show memiliki lebih dari 6 season, yang menunjukkan adanya konten dengan tingkat retensi tinggi dan potensi long-term engagement pada pengguna.





Insight & Business

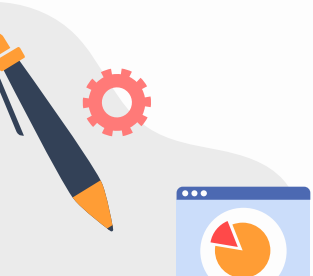
<u>Konten seperti apa yang paling dominan?</u>	Konten Movie mendominasi, menunjukkan preferensi konsumsi cepat oleh user.
<u>Apakah ada tren peningkatan konten dari tahun ke tahun?</u>	Terdapat tren peningkatan konten dari tahun ke tahun
<u>Genre apa yang paling potensial?</u>	Genre potensial adalah drama dan internasional Movies, dengan target market utama United States
<u>Negara mana yang layak ditargetkan?</u>	United States dengan total konten 2,819 menjadi target yang layak





Berikan 3 Rekomendasi strategis

- 01** Fokus pada produksi film berdurasi 90–120 menit untuk meningkatkan engagement user
- 02** Ekspansi ke negara dengan demand tinggi seperti United States untuk memperluas market
- 03** Produksi konten dengan rating TV-MA / TV-14 untuk menjangkau segmen audiens terbesar





SQL Challenge

```
SELECT release_year,  
AVG(duration_minutes) as avg_duration  
FROM `netflix_ds.analysis_view`  
WHERE type = 'Movie'  
GROUP BY release_year  
ORDER BY release_year DESC  
LIMIT 5;
```

Row	release_year	avg_duration
1	2021	96.44404332129...
2	2020	92.14119922630...
3	2019	93.46603475513...
4	2018	96.18513689700...
5	2017	95.53524804177...

```
SELECT country, COUNT(*) AS total  
FROM `netflix_ds.analysis_view`  
GROUP BY country  
ORDER BY total DESC  
LIMIT 5
```

Row	country	total
1	United States	2819
2	India	972
3	Unknown	831
4	United Kingdom	421
5	Japan	245





Netflix menunjukkan dominasi Movie, pertumbuhan sejak 2015, dan distribusi global yang terpusat.

Pendekatan berbasis data ini dapat membantu perusahaan dalam menentukan strategi produksi dan ekspansi konten secara lebih efektif.